Learning Objective 3.5: I can implement beam steering on the ADALM-PHASER and verify the steered pattern against theory.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **Building the phase table for a new angle.** Your HB100 locks at 10.525 GHz and the PHASER array has $N = 8$ elements on a $d = 14$ mm pitch. You plan to steer to $\theta_0 = 20°$.

   (a) Find the wavelength and the electrical spacing d/λ.

   $$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{10.525 \times 10^9} = 0.02850 \text{ m} = 28.50 \text{ mm}$$

   $\lambda =$ | 28.5 mm |

   $$\frac{d}{\lambda} = \frac{14}{28.50} = 0.491$$

   $d/\lambda =$ | 0.491 |

   (b) Compute the progressive element phase $\Delta\phi$ required to steer the beam to $20°$.

   $\Delta\phi =$ | $60.5°$ |

   $$\Delta\phi = kd\sin(\theta_0) = 360° \frac{d}{\lambda} \sin(\theta_0)$$

   $$= 360°(0.491)(0.3420) = 176.8°(0.3420) = 60.5°$$

   (c) Complete the table of phases the beamformer will apply, wrapping each value into $0°$ to $360°$. State which element is the first to wrap.

Element	1	2	3	4	5	6	7	8
Applied phase								

   Element n carries $(n-1)\Delta\phi$ with $\Delta\phi = 60.5°$, wrapped modulo $360°$:

   $$0.0°,\ 60.5°,\ 121.0°,\ 181.4°,\ 241.9°,\ 302.4°,\ 2.9°,\ 63.4°$$

   Element 7 is the first to wrap: its unwrapped value is $6(60.5°) = 362.9°$, which the shifter produces as $2.9°$.

   (d) A classmate says the wrapped table cannot produce the same beam as the unwrapped ramp, because element 7 now leads element 1 by only $2.9°$ instead of lagging a full turn behind. In one or two sentences, explain why the two tables give identical patterns.

   The field radiated by an element depends on its phase only through $e^{j\phi}$, which is unchanged by adding or subtracting any whole number of $360°$. A $362.9°$ shift and a $2.9°$ shift are the same complex weight, so every term in the array sum is identical and the patterns are identical. Wrapping matters only because the hardware cannot store a number larger than $360°$.

2. **Reading a measured sweep.** A cadet places the HB100 at $+30.0^\circ$ on the protractor arc, hunts for the peak in the Beam Steering box, and then runs a full sweep on the Rectangular tab. The trace peaks at a commanded angle of 28.1°, its half-power crossings are at 20.5° and 35.8°, the peak sits 0.7 dB below the frozen boresight trace, and the first sidelobes read -11.8 dBc. The steer resolution is the default 2.8125°.

   (a) Find the measured half-power beamwidth and compare it with the value predicted from the 13.1° boresight beamwidth.

   θ_{HP} = 15.3°

   $$\theta_{HP,meas} = 35.8^\circ - 20.5^\circ = 15.3^\circ$$

   The $1/\cos(\theta_0)$ broadening predicts

   predicted = 15.1°

   $$\frac{13.1^\circ}{\cos(30^\circ)} = \frac{13.1^\circ}{0.866} = 15.1^\circ$$

   The measurement is 0.2° wide, well inside one grid step.

   (b) Express the difference between the physical angle and the commanded angle at the peak as a fraction of one sweep grid step.

   offset = 0.68 steps

   $$\Delta\theta = 30.0^\circ - 28.1^\circ = 1.9^\circ$$

   $$\frac{1.9^\circ}{2.8125^\circ} = 0.68 \text{ grid steps}$$

   (c) The cadet reports the array as broken because the beam did not go to 30°. In two or three sentences, say whether the data support that conclusion.

   They do not. The sweep only samples commanded angles on a 2.8125° grid, so the reported peak is the grid point nearest the true peak and can be wrong by half a step, 1.41°, on the grid alone. Adding the protractor and aiming error of roughly 1.5° puts a 1.9° disagreement inside the expected spread. A pointing fault would show up as an offset of several degrees that persists at every commanded angle.

   (d) The peak fell 0.7 dB when the beam was steered to 30°, while the projected-aperture rule predicts a drop of $10\log_{10}(\cos(30^\circ)) = -0.6$ dB instead. Give the most likely reason the measured loss is the larger of the two, and state whether you would expect the gap to grow or shrink at 50°.

   The projected-aperture rule is the ideal-element bound: it assumes every patch radiates equally in all scan directions and charges only for the foreshortened aperture. A real patch element has its own pattern that rolls off away from broadside, closer to $\cos^{1.3}\theta$ to $\cos^{1.5}\theta$ in power, so the measured scan loss is always at least as large as the bound. The gap grows with scan angle, so a 50° scan would show a larger excess than the 0.1 dB seen here.

3. **Bounding the error sources.** You need to state an uncertainty for the peak angle you measured. Work with the default 2.8125° steer resolution, a source range of 1.00 m, and an HB100 nominal frequency of 10.525 GHz.

   (a) From the sweep grid alone, give the worst-case error in the peak angle read off the trace, and the RMS error if the true peak is equally likely to fall anywhere between two grid points.

   worst case = 1.41°

   RMS = 0.81°

   The reported peak is the nearest grid point, so the worst case is half a step, and a uniform spread of width Δ has standard deviation $\Delta/\sqrt{12}$:

$$\Delta\theta_{\text{max}} = \frac{2.8125^\circ}{2} = 1.41^\circ$$
$$\Delta\theta_{\text{rms}} = \frac{2.8125^\circ}{\sqrt{12}} = 0.81^\circ$$

   (b) The phases were computed for 10.525 GHz and applied for $\theta_0 = 45^\circ$. The module then drifts to 10.325 GHz. Find the angle the beam points at, and the resulting error.

   θ = 46.1°

   $\Delta\theta$ = $+1.1^\circ$

   The ramp is fixed, so the beam sits where $kd\sin(\theta) = \Delta\phi$ at the new frequency:

$$\theta = \arcsin\left(\frac{f_0}{f}\sin(\theta_0)\right) = \arcsin\left(\frac{10.525}{10.325}(0.7071)\right)$$
$$= \arcsin(0.7208) = 46.1^\circ, \qquad \Delta\theta = +1.1^\circ$$

   (c) How far along the arc must the tripod be misplaced to introduce a 1.0° error in the reference angle?

   s = 17 mm

$$s = R\,\theta_{\text{rad}} = (1.00\text{ m})\left(\frac{1.0\pi}{180}\right) = 0.0175\text{ m}$$

   which is 17 mm of lateral placement error.

   (d) Multipath in the classroom adds roughly 1° of peak-angle wander. Combine all four contributions in quadrature, then say which one you could reduce from the GUI and how, and which one you could not.

   Taking 1.41° (grid), 1.5° (aim), 1.1° (drift) and 1.0° (multipath):

$$\sqrt{1.41^2 + 1.5^2 + 1.1^2 + 1.0^2} = 2.5^\circ$$

   The grid term is the one the GUI controls: a smaller Steer Resolution, or more Phase Shift Bits, samples the trace more finely. Drift cannot be fixed from the GUI, since the HB100 free-runs with no reference input.

4. **Working backwards from the phase read-back.** A lab partner recorded the Phase Control values below but forgot which steer angle was commanded. The HB100 was at 10.525 GHz.

Element	1	2	3	4	5	6	7	8
Applied phase	0.0°	125.0°	250.1°	15.1°	140.1°	265.2°	30.2°	155.2°

(a) Recover the progressive phase $\Delta\phi$ from the table.

Take successive differences and add 360° wherever the value drops:

$$125.0^\circ,\ 125.1^\circ,\ (375.1 - 250.1) = 125.0^\circ,\ \ldots$$

Every step is the same, so

$$\Delta\phi = 125.0^\circ$$

$\Delta\phi$ = 125.0°

(b) Find the commanded steer angle.

$$\sin(\theta_0) = \frac{\Delta\phi}{360^\circ(d/\lambda)} = \frac{125.0^\circ}{176.8^\circ} = 0.7071$$

$$\theta_0 = 45.0^\circ$$

θ_0 = 45.0°

(c) The same read-back is also consistent with $\Delta\phi = 125.0^\circ + 360^\circ = 485.0^\circ$. Explain in one or two sentences why that second solution can be discarded for this array.

A progressive phase of 485.0° would require $\sin(\theta_0) = 485.0/176.8 = 2.74$, which no real angle satisfies. Because $d/\lambda = 0.491$ is less than one half, every wrapped phase table maps to exactly one beam direction in visible space, and the ambiguity does not arise. Spacings above a half wavelength are where the second solution becomes real, and that is the grating lobe of L23.

(d) The partner later finds the HB100 was running at 10.300 GHz while these phases were applied. Where did the beam point?

At 10.300 GHz, $\lambda = 29.13$ mm and $d/\lambda = 0.4807$, so

$$\sin(\theta_0) = \frac{125.0^\circ}{360^\circ(0.4807)} = \frac{125.0^\circ}{173.0^\circ} = 0.7226$$

$$\theta_0 = 46.3^\circ$$

The beam sat 1.3° past the commanded angle.

θ_0 = 46.3°

5. **Two frozen traces.** A group freezes the boresight sweep, then steers to 45° and sweeps again. The boresight trace has a half-power width of 13.1° and defines the 0 dB reference. The steered trace peaks at 45.0°, is 18.9° wide at half power, and its peak is 1.7 dB below the boresight peak.

   (a) Predict the half-power width at 45° and compare it with the measurement.

   θ_{HP} = 18.5°

   The beam broadens as $1/\cos(\theta_0)$, so scaling the measured 13.1° broadside width gives

   $$\theta_{\text{HP}} \approx \frac{13.1^\circ}{\cos(45^\circ)} = \frac{13.1^\circ}{0.7071} = 18.5^\circ$$

   The trace reads 18.9°, which is 0.4° wide and inside one grid step. Starting instead from the theory width $0.886\lambda/(Nd) = 13.2^\circ$ the same rule gives 18.7°, so either chain lands inside one grid step of the trace.

   (b) Predict the peak drop from the projected-aperture rule and compare it with the measured 1.7 dB drop.

   ΔG = -1.5 dB

   $$\begin{aligned}\Delta G &= 10\log_{10}(\cos(45^\circ)) \\ &= 10\log_{10}(0.7071) = -1.5 \text{ dB}\end{aligned}$$

   The measurement is 0.2 dB deeper, consistent with the element pattern rolling off faster than the ideal cosine.

   (c) In one or two sentences, name the single geometric fact behind both the wider beam and the lower peak.

   Steering tilts the beam off the face of the aperture, so the array presents a projected length of $Nd\cos(\theta_0)$ rather than Nd to the incoming wave. A shorter effective aperture gives a proportionally wider beam and collects proportionally less power, which is the same $\cos(\theta_0)$ appearing in both results.

   (d) A radar specification calls for a 13° beam maintained out to 45° of scan. How many elements would this array need at the same 14 mm spacing, and state the cost of that choice in one phrase.

   N = 12 elements

   The projected aperture must equal the present one at 45°:

   $$N\cos(45^\circ) = 8$$
   $$N = \frac{8}{0.7071} = 11.3 \rightarrow 12 \text{ elements}$$

   The cost is hardware: twelve elements need twelve LNAs and phase shifters, three ADAR1000 chips instead of two, and a wider aperture that still loses the same 1.5 dB of scan loss.

Documentation: _____